

MOHAMMAD MEHDI ASGARI

+98 916 664 8219 | Mohammadmehdi.asgari262@gmail.com | 0009-0000-9528-0970

https://mohammad-mehdi-asgari.github.io

Education

2018-2022	M.Sc. of Seafood Processing / Sari Agricultural Sciences and Natural Resources University GPA: 16.48/20.00
2014-2018	B.Sc. of Natural Science Engineering - Fisheries / Sari Agricultural Sciences and Natural Resources University Majoring in Aquaculture Breeding and Culture

Research Experience

Islamic Azad University,
Qaemshahr Branch
2025- Ongoing

The Effect of Different Levels of *Ganoderma lucidum* in the Diet on Growth Indices, Non-specific Immunity and Antioxidant Enzyme Activity in Siberian sturgeon (*Acipenser baerii*) / Advisor & Experimental Designer

- Successfully executed a 60-day feeding trial by incorporating three different concentrations of *Ganoderma lucidum* powder.
- Conducted comprehensive **in vivo sampling** at the end of the trial, including **blood collection** and **liver sampling**, to assess **key health and performance markers**.
- Determined standard **growth performance** indices (e.g., WG, SGR, FCR), **Immunity Markers Analyzed** (such as Lysozyme activity, Complement system activity), **Antioxidant Enzyme Assays** (Such as Assessed activities of Catalase (CAT) and Superoxide Dismutase).

University of Tehran
2024 - Ongoing

Combined Effects of Astaxanthin, Vitamin E, and Selenium as Anti-Stress Agents under High-Temperature Stress in Beluga (*Huso huso*) / Advisor & Experimental Designer

- I Designed **16** independent Ras systems for experiments, focusing on wide range of Aquaculture studies such as nutrition and digestion, diseases, immunity and stress studies.
- we examined **growth indices** (such as FW, WG); **stress indices** (e.g., ammonia, cortisol); and **immunology indices** (such as Serum lysozyme activity and Immunoglobulin).
- Assisted with fish acclimation, maintenance, troubleshooting during experiments and data analysis.

University of Tehran
2023 - 2025

Combined Effects of the Anti-Stress Agents Arginine, Vitamin C, and Zinc on Hypoxia-Induced Stress in Rainbow Trout (*Oncorhynchus mykiss*) / Advisor & Experimental Designer

- Designed **12** independent **RAS systems** for this experiments and wide range of aquaculture studies.
- Three treatments (T) with three different concentrations of anti-stressor and we examined **growth indices** (such as WG, SGR); **stress indices** (e.g., glucose level and total antioxidants); and **immunology indices** (such as peroxidase and ACH50 level).
- Assisted with fish acclimation, maintenance, troubleshooting during experiments and data analysis.
- Results** : No significant difference in all treatments in growth indices, T2 with the average level of anti-stressors in stress indices and T1 with the lowest level of anti-stressors in immune indices had better results than the others in terms of performance and economy.

University of Tehran
2023 - 2025

Investigation of the Feasibility of Producing Hydrolyzed Collagen from Nile Tilapia (*Oreochromis niloticus*) Skin Using a Microbial Method / Advisor & Experimental Designer

- Extracted collagen from Nile tilapia skin and hydrolysed it using *Trichoderma ghanense* (DH ≈ 27%).
 - Structural analyses (**FT-IR, DSC**) confirmed peptide formation and enhanced thermal stability.
- Result:** Developed hydrolyzed collagen from Nile tilapia skin through microbial fermentation, resulting in low-molecular-weight peptides, improved solubility, and potential functional applications.

University of Tehran
2025 - 2026

Ultrasound Thawing Enhances Myofibrillar Protein Integrity and Quality Attributes of Common Carp (*Cyprinus carpio*) Fillets

<https://doi.org/10.1002/fsn3.71833>

- Applied three thawing methods (Ultrasound, Water Immersion, Microwave)
- Analyzed effects on fillet and protein structure using various analytical techniques (such as **SEM, TPA, Color, DLS, Zeta Potential, SDS-Page**).

Results: Found ultrasound most effective in preserving fillet and protein structure.

Research Interest

Aquatic Animal Health / Fish Immunology and Host-Microbiome Interactions / Fish Nutrition and Diet-Based Health Modulation / Aquaculture Stress Physiology and Disease Response

Academic Project

Oct 2025-Current

- Ecological Study on Rainbow Trout (*Oncorhynchus mykiss*) & Brown Trout (*Salmo trutta*)**
A study on the population, morphology, behavior, feeding habits, and environmental quality parameters of Rainbow Trout , Brown Trout and species diversity in Tar and Lar Lake

Nov 2024- Nov 2025

- Ecological Study on Pike (*Esox lucius*)**
A study on the population, morphology, behavior, feeding habits, and environmental quality parameters of Pike (*Esox lucius*) in Ovan Lake

Oct 2023 - Sep 2024

- Ecological Study on Brown Trout (*Salmo trutta*)**
A study on the population, morphology, behavior, feeding habits, and environmental quality parameters of Brown Trout in Chalus River

WORK
Experience

University of Tehran
Oct 2023 - Current

SnappFood
June 2020 - May 2023

Nilofar Abi Co.
Feb 2018 - May 2020

Behbudi Aqua
Jun2015-Jan2018

Research and Lab Assistant

- Led 10+ field visits to farms and seafood plants; boosted student engagement by 40%.
- Taught 3 courses in quality control and aquaculture to 80+ students; >90% satisfaction.
- Supervised 4+ M.Sc. and PhD thesis in aquaculture technology and seafood quality.
- Established a reproduction and feeding-trial facility; increased research capacity by 35%.
- Supported lab operations (fisheries and seafood), equipment upkeep, and academic coordination.

Food Quality Controller

- Real-time issue resolution (200+ orders/day)
- Quality & hygiene audits (95% compliance)
- Onboarding & evaluating 20+ restaurants
- Cross-team coordination (restaurants-logistics-support)
- Complaint rate reduction (~25%)
- Food quality control in digital services
- Customer experience monitoring
- Operational performance assessment
- Strong analytical & communication skills

Fish Farm Supervisor

- Managed daily operations across four cold-water ponds producing 30+ tons of salmonids.
- Led a team of 10 technicians, improving operational efficiency by ~25%.
- Monitored water quality, feeding schedules, and health status, keeping survival rates above 90%.
- Reduced early-life mortality of salmonids to below 11%.
- Improved fillet quality (texture, colour, nutritional profile) by an average of 6% annually.
- Supported sales planning and client relations, contributing to a 15% increase in customer retention.
- Gained hands-on expertise in broodstock management, aquaculture production cycles, and post-harvest quality control

AQUACULTURE TECHNICIAN → HEAD OF FISH BREEDING AND HOLDING FACILITY

- Promoted from technician to department head based on performance and leadership
- Designed 100+ saltwater RAS aquarium systems, improving efficiency ~20%
- Monitored coral and fish sections; maintained water-quality parameters within ±3% of optimal levels
- Supervised broodstock health of 150+ marine species, improving hatch success by 15% annually
- Diagnosed and treated 30+ disease cases, reducing mortality by 25%
- Oversaw quarantine operations for 100+ monthly imports with zero cross-contamination
- Delivered 12+ staff training sessions annually, improving team performance by 18%

Teacher
Assistant

University of Tehran
Oct 2023 - Current

University of Tehran
Oct 2023 - Current

University of Tehran
Oct 2023 - Current

“Cold-Water Fish Breeding” – Delivered full lectures and practical sessions to undergraduate students

“Breeding and Rearing of Ornamental Fish” – Delivered full lectures and practical sessions to undergraduate students

“Quality Control of Fishery Products” – Delivered full lectures and practical sessions to undergraduate students

Skills

Hard Skills

Laboratory Techniques & Experimental Design, AFM, SEM, Python, SPSS, HTML, Written & Verbal Communication Skills

Soft Skills

Communication & Presentation, Leadership & Teaching, Adaptability & Time Management, Teamwork & Collaboration, Problem Solving & Critical Thinking

Selected
Conferences

May 2025

Dec 2023

Nov 2022

Feb 2022

Nov 2020

Honours&
Awards

- 1st National Conference on Technology and innovation in aquaculture, environment and food security
- 7th International Congress of Fisheries and Aquatic Research
- Sixth International Congress of Fisheries and Aquatic Research
- 1st National Conference on Modern Researches in Biological Sciences
- Fourth International Fisheries and Aquatic Research Congress

- Ranked Top %1 Among Participants in The Undergraduate Nationwide Entrance Exam
- Ranked 114th Among +4500 Participants in Master Nationwide Entrance Exam
- Ranked 1st Among Fisheries Engineering, Seafood Processing Students

Language

Language: Persian – Native / English – IELTS 7.5